

Lie Algebras and Matrix Lie Algebras

Road map

The local, infinitesimal structure of a Lie group is encoded by a Lie algebra. This chapter first defines Lie algebras abstractly, derives the component form of antisymmetry and the Jacobi identity, and explains how structure constants determine the bracket. It then develops subalgebras, homomorphisms, and the fundamental matrix example $\mathfrak{gl}(V)$. By differentiating the defining equations of matrix groups at the identity, we obtain

$$\mathfrak{so}(n), \quad \mathfrak{sp}(2n, \mathbb{R}), \quad \mathfrak{u}(n), \quad \mathfrak{su}(n), \quad \mathfrak{sl}(n, \mathbb{K}),$$

including their dimensions and useful bases. A table collects all group–algebra pairs from the lecture. The final sections introduce the derived algebra and derive the precise transformation rule of the structure constants under a basis change.

14.1 From a Lie group to its infinitesimal algebra

The basic correspondence has two directions:

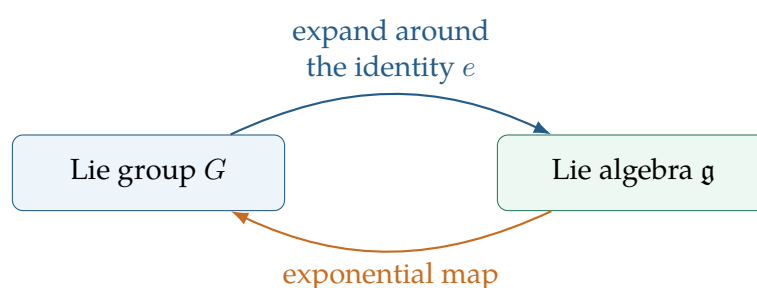


图 14.1: The local Lie correspondence. This chapter constructs and studies the algebra side; the exponential map and the global correspondence are developed in the following lecture.

For matrix groups, “expand around e ” means taking derivatives of smooth curves through the identity matrix. The result is a vector space with a bilinear product measuring noncommutativity. We first axiomatize that structure independently of any group.

14.2 Definition and immediate consequences

Definition 14.1: Lie algebra

Let V be a d -dimensional vector space over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. A *Lie bracket* on V is a map

$$[\cdot, \cdot] : V \times V \longrightarrow V, \quad (X, Y) \longmapsto [X, Y],$$

such that, for all $X, Y, Z \in V$ and $\alpha, \beta \in \mathbb{K}$:

1. **Bilinearity:**

$$[\alpha X + \beta Y, Z] = \alpha[X, Z] + \beta[Y, Z],$$

$$[X, \alpha Y + \beta Z] = \alpha[X, Y] + \beta[X, Z].$$

2. **Antisymmetry:**

$$[X, Y] = -[Y, X].$$

3. **Jacobi identity:**

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0.$$

The pair $\mathfrak{g} = (V, [\cdot, \cdot])$ is a real or complex *Lie algebra*, and $\dim \mathfrak{g} = \dim V = d$.

Several identities follow at once:

$$[X, X] = -[X, X] \implies [X, X] = 0,$$

because the fields $\mathbb{R}$ and $\mathbb{C}$ have characteristic zero. Bilinearity in the second argument follows from bilinearity in the first together with antisymmetry, and

$$[X, 0] = [X, Y - Y] = [X, Y] - [X, Y] = 0.$$

Definition 14.2: Abelian Lie algebra

A Lie algebra $\mathfrak{g}$ is *Abelian* if

$$[X, Y] = 0 \quad \text{for every } X, Y \in \mathfrak{g}.$$

14.3 Generators and structure constants

Let

$$\{T_1, \dots, T_d\}$$

be a basis of $\mathfrak{g}$. In the terminology of the lecture, the basis vectors T_i are *generators*. Each bracket of basis vectors has a unique expansion

$$[T_i, T_j] = \sum_{k=1}^d f_{ij}^k T_k. \quad (14.1)$$

The scalars $f_{ij}^k \in \mathbb{K}$ are the *structure constants* of $\mathfrak{g}$ with respect to this basis.

Antisymmetry gives

$$\sum_k f^k_{ij} T_k = [T_i, T_j] = -[T_j, T_i] = -\sum_k f^k_{ji} T_k.$$

Uniqueness of basis coefficients implies

$$f^k_{ij} = -f^k_{ji}. \quad (14.2)$$

Now impose the Jacobi identity on T_i, T_j, T_k :

$$\begin{aligned} 0 &= [T_i, [T_j, T_k]] + [T_j, [T_k, T_i]] + [T_k, [T_i, T_j]] \\ &= \sum_\ell f^\ell_{jk} [T_i, T_\ell] + \sum_\ell f^\ell_{ki} [T_j, T_\ell] + \sum_\ell f^\ell_{ij} [T_k, T_\ell] \\ &= \sum_{\ell, m} \left(f^\ell_{jk} f^m_{i\ell} + f^\ell_{ki} f^m_{j\ell} + f^\ell_{ij} f^m_{k\ell} \right) T_m. \end{aligned}$$

Therefore the component Jacobi constraints are

$$\boxed{\sum_{\ell=1}^d \left(f^\ell_{jk} f^m_{i\ell} + f^\ell_{ki} f^m_{j\ell} + f^\ell_{ij} f^m_{k\ell} \right) = 0} \quad (14.3)$$

for every i, j, k, m .

Conversely, suppose a d -dimensional vector space has a basis $\{T_i\}$, and scalars f^k_{ij} satisfy (14.2) and (14.3). For

$$X = \sum_i x_i T_i, \quad Y = \sum_j y_j T_j,$$

define

$$[X, Y] = \sum_{i,j,k} x_i y_j f^k_{ij} T_k. \quad (14.4)$$

Bilinearity is built into the formula, antisymmetry follows from (14.2), and the Jacobi identity follows from (14.3). Thus suitable structure constants completely determine a Lie algebra.

14.4 First examples

14.4.1 Every one-dimensional Lie algebra is Abelian

Let $\dim \mathfrak{g} = 1$ and choose a basis $\{T\}$. There is a scalar f with

$$[T, T] = fT.$$

But $[T, T] = 0$, so $f = 0$. If $X = xT$ and $Y = yT$, then

$$[X, Y] = xy[T, T] = 0.$$

Hence every Lie bracket on a one-dimensional vector space is Abelian.

14.4.2 $\mathbb{R}^3$ with the cross product

Let

$$\mathfrak{g} = (\mathbb{R}^3, \times), \quad [X, Y] = X \times Y.$$

The cross product is bilinear and antisymmetric. Its Jacobi identity can be checked with the vector triple-product identity

$$A \times (B \times C) = B(A \cdot C) - C(A \cdot B).$$

Indeed,

$$\begin{aligned} X \times (Y \times Z) + Y \times (Z \times X) + Z \times (X \times Y) \\ = Y(X \cdot Z) - Z(X \cdot Y) \\ + Z(Y \cdot X) - X(Y \cdot Z) \\ + X(Z \cdot Y) - Y(Z \cdot X) = 0. \end{aligned}$$

For the canonical basis e_1, e_2, e_3 ,

$$[e_i, e_j] = e_i \times e_j = \sum_k \varepsilon_{ijk} e_k.$$

Thus the structure constants are the Levi-Civita symbols:

$$f^k_{ij} = \varepsilon_{ijk}.$$

14.4.3 The general linear Lie algebra

Example 14.1: Matrix general linear algebra

$$\mathfrak{gl}(n, \mathbb{K}) = M_n(\mathbb{K})$$

becomes a Lie algebra under the commutator

$$[X, Y] = XY - YX.$$

Antisymmetry is immediate. For the Jacobi identity, expand:

$$\begin{aligned} [X, [Y, Z]] &= XYZ - XZY - YZX + ZYX, \\ [Y, [Z, X]] &= YZX - YXZ - ZXY + XZY, \\ [Z, [X, Y]] &= ZXY - ZYX - XYZ + YXZ. \end{aligned}$$

Every monomial cancels, so the sum is zero.

For an n -dimensional vector space V , define

$$\mathfrak{gl}(V) = \text{End}_{\mathbb{K}}(V)$$

with

$$[f_1, f_2] = f_1 \circ f_2 - f_2 \circ f_1.$$

This is the *general linear Lie algebra of degree n* . Its dimension is

$$\dim_{\mathbb{K}} \mathfrak{gl}(V) = n^2.$$

If $v_1, \dots, v_n$ is a basis of V , let E_{ij} be the matrix unit that sends v_j to v_i and annihilates the other basis vectors. Then $\{E_{ij}\}_{1 \leq i, j \leq n}$ is a basis of $\mathfrak{gl}(V)$, and

$$[E_{ij}, E_{kl}] = \delta_{jk} E_{il} - \delta_{li} E_{kj}. \quad (14.5)$$

14.5 Subalgebras and homomorphisms

Definition 14.3: Lie subalgebra

A Lie subalgebra $\mathfrak{h} \leq \mathfrak{g}$ is a linear subspace closed under the bracket:

$$X, Y \in \mathfrak{h} \implies [X, Y] \in \mathfrak{h}.$$

The whole algebra $\mathfrak{g}$ and the zero algebra $\{0\}$ are the trivial Lie subalgebras.

Definition 14.4: Lie-algebra homomorphism and isomorphism

For Lie algebras $\mathfrak{g}, \mathfrak{g}'$ over the same field, a map $\psi : \mathfrak{g} \rightarrow \mathfrak{g}'$ is a *homomorphism* if it is linear and

$$\psi([X, Y]) = [\psi(X), \psi(Y)] \quad \text{for all } X, Y \in \mathfrak{g}.$$

A bijective homomorphism is an *isomorphism*. If an isomorphism exists, we write

$$\mathfrak{g} \cong \mathfrak{g}'.$$

Example 14.2: Matrices of endomorphisms

Choose a basis $v_1, \dots, v_n$ of V . If

$$f(v_j) = \sum_i A_{ij} v_i,$$

let $\psi(f) = A$. Then

$$\psi : \mathfrak{gl}(V) \longrightarrow \mathfrak{gl}(n, \mathbb{K})$$

is linear and bijective. Since matrix multiplication represents composition,

$$\psi([f_1, f_2]) = \psi(f_1 \circ f_2 - f_2 \circ f_1) = \psi(f_1)\psi(f_2) - \psi(f_2)\psi(f_1),$$

so

$$\mathfrak{gl}(V) \cong \mathfrak{gl}(n, \mathbb{K}).$$

14.5.1 $(\mathbb{R}^3, \times) \cong \mathfrak{so}(3)$

Define

$$T_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad T_y = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad T_z = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Direct multiplication gives

$$[T_i, T_j] = \sum_k \varepsilon_{ijk} T_k.$$

Their span is

$$\text{span}_{\mathbb{R}}\{T_x, T_y, T_z\} = \left\{ \begin{pmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{pmatrix} : x_1, x_2, x_3 \in \mathbb{R} \right\} = \mathfrak{so}(3).$$

The map

$$\psi : \mathbb{R}^3 \longrightarrow \mathfrak{so}(3), \quad (x_1, x_2, x_3) \longmapsto x_1 T_x + x_2 T_y + x_3 T_z,$$

is linear and bijective. On basis elements,

$$\psi([e_i, e_j]) = \sum_k \varepsilon_{ijk} \psi(e_k) = \sum_k \varepsilon_{ijk} T_k = [T_i, T_j].$$

Bilinearity extends this equality to all X, Y , hence

$$(\mathbb{R}^3, \times) \cong \mathfrak{so}(3). \quad (14.6)$$

Theorem 14.1: Ado's theorem

Every finite-dimensional Lie algebra over $\mathbb{R}$ or $\mathbb{C}$ is isomorphic to a Lie subalgebra of $\mathfrak{gl}(n, \mathbb{K})$ for some n .

The lecture states Ado's theorem without proof. It justifies concentrating on matrix, or *linear*, Lie algebras when studying finite-dimensional Lie algebras.

14.6 Extracting a Lie algebra from a matrix Lie group

Let $G \leq GL(N, \mathbb{K})$ be a matrix Lie group. Consider all smooth curves

$$A : (-\epsilon, \epsilon) \longrightarrow G, \quad A(0) = I.$$

Their velocity vectors at the identity form the tangent space

$$T_I G = \{\dot{A}(0) : A(t) \in G, A(0) = I\}.$$

For matrix Lie groups, this tangent space is the associated Lie algebra $\mathfrak{g}$. Its bracket is the matrix commutator. In the next chapter we justify this construction and its connection with one-parameter subgroups; here we use it to derive examples directly.

14.6.1 The orthogonal and special orthogonal algebras

Let $A(t) \in O(n)$ and $A(0) = I$. Since

$$A(t)^T A(t) = I,$$

write

$$A(t) = I + tX + O(t^2), \quad X = \dot{A}(0).$$

Then

$$\begin{aligned} I &= (I + tX^T + O(t^2))(I + tX + O(t^2)) \\ &= I + t(X^T + X) + O(t^2). \end{aligned}$$

Therefore $X^T + X = 0$. This gives

$$\mathfrak{o}(n) = \mathfrak{so}(n) = \{X \in M_n(\mathbb{R}) : X^T = -X\}. \quad (14.7)$$

It is a vector subspace and is closed under commutators, as proved explicitly in the gauge-theory chapter. The independent entries lie above the diagonal, so

$$\dim \mathfrak{so}(n) = \frac{1}{2}n(n-1).$$

Every skew-symmetric matrix has zero trace. Thus the determinant-one condition defining $SO(n)$ imposes no further infinitesimal constraint, which explains why $O(n)$ and $SO(n)$ have the same Lie algebra. For $n = 3$, the basis T_x, T_y, T_z above makes the comparison with $(\mathbb{R}^3, \times)$ explicit.

14.6.2 Upper triangular matrices

The group

$$UT_n(\mathbb{K}) = \{A \in GL(n, \mathbb{K}) : A_{ij} = 0 \text{ for } i > j\}$$

has Lie algebra

$$\mathfrak{ut}_n(\mathbb{K}) = \{X \in M_n(\mathbb{K}) : X_{ij} = 0 \text{ for } i > j\}.$$

Products and commutators of upper triangular matrices remain upper triangular. A basis is

$$\{E_{ij} : 1 \leq i \leq j \leq n\},$$

and

$$\dim_{\mathbb{K}} \mathfrak{ut}_n(\mathbb{K}) = \frac{1}{2}n(n+1).$$

14.7 The symplectic Lie algebra

Recall

$$Sp(2n, \mathbb{R}) = \{A \in GL(2n, \mathbb{R}) : A^T J A = J\}, \quad J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

For a path $A(t) = I + tX + O(t^2)$ through the identity,

$$\begin{aligned} J &= A(t)^T J A(t) \\ &= (I + tX^T + O(t^2))J(I + tX + O(t^2)) \\ &= J + t(X^T J + JX) + O(t^2). \end{aligned}$$

Thus

$$\mathfrak{sp}(2n, \mathbb{R}) = \{X \in M_{2n}(\mathbb{R}) : X^T J + JX = 0\}. \quad (14.8)$$

Closure under the commutator can be verified exactly as in the notes. If $X_1^T J = -JX_1$ and $X_2^T J = -JX_2$, then

$$\begin{aligned} [X_1, X_2]^T J &= (X_2^T X_1^T - X_1^T X_2^T)J \\ &= X_2^T(-JX_1) - X_1^T(-JX_2) \\ &= JX_2X_1 - JX_1X_2 \\ &= -J[X_1, X_2]. \end{aligned}$$

Hence $[X_1, X_2] \in \mathfrak{sp}(2n, \mathbb{R})$.

14.7.1 Block form and dimension

Write

$$X = \begin{pmatrix} A & B \\ C & D \end{pmatrix}.$$

Then

$$\begin{aligned} X^T J + JX &= \begin{pmatrix} A^T & C^T \\ B^T & D^T \end{pmatrix} \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} + \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} \\ &= \begin{pmatrix} -C^T & A^T \\ -D^T & B^T \end{pmatrix} + \begin{pmatrix} C & D \\ -A & -B \end{pmatrix} \\ &= \begin{pmatrix} C - C^T & A^T + D \\ -D^T - A & B^T - B \end{pmatrix}. \end{aligned}$$

The condition is therefore equivalent to

$$B = B^T, \quad C = C^T, \quad D = -A^T.$$

The free block A contributes n^2 real parameters, and each symmetric block contributes $n(n+1)/2$. Hence

$$\dim \mathfrak{sp}(2n, \mathbb{R}) = n^2 + 2 \frac{n(n+1)}{2} = n(2n+1). \quad (14.9)$$

14.7.2 The rank-one case and a correction

For $n = 1$, the condition is simply $\text{tr } X = 0$, so

$$\mathfrak{sp}(2, \mathbb{R}) = \mathfrak{sl}(2, \mathbb{R}).$$

A basis used in the handwritten notes is

$$T_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad T_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad T_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Direct calculation gives

$$[T_1, T_2] = 2T_3, \quad [T_2, T_3] = 2T_1, \quad [T_3, T_1] = -2T_2. \quad (14.10)$$

Correction to the handwritten $\mathfrak{sp}(2, \mathbb{R})$ comparison

The notes write $[T_i, T_j] = 2 \sum_k \varepsilon_{ijk} T_k$, rescale $T'_i = \frac{1}{2} T_i$, and propose

$$T'_1 \mapsto T_x, \quad T'_2 \mapsto T_y, \quad T'_3 \mapsto T_z$$

as an isomorphism with $\mathfrak{so}(3)$. Equation (14.10) shows the missing sign:

$$[T'_3, T'_1] = -T'_2, \quad [T_z, T_x] = T_y.$$

Therefore the proposed map does not preserve the bracket. Without rescaling, the factor also fails:

$$\psi([T_1, T_2]) = 2T_z \neq [T_x, T_y] = T_z.$$

The correct real statement is

$$\boxed{\mathfrak{sp}(2, \mathbb{R}) \cong \mathfrak{sl}(2, \mathbb{R}) \not\cong \mathfrak{so}(3).}$$

The two three-dimensional real algebras are different real forms; after complexification one has

$$\mathfrak{sp}(2, \mathbb{C}) \cong \mathfrak{sl}(2, \mathbb{C}) \cong \mathfrak{so}(3, \mathbb{C}).$$

14.8 Unitary and special unitary Lie algebras

Let

$$SU(n) = \{A \in GL(n, \mathbb{C}) : A^\dagger A = I, \det A = 1\}.$$

For a path $A(t) = I + tX + O(t^2)$,

$$\begin{aligned} I &= A(t)^\dagger A(t) \\ &= (I + tX^\dagger + O(t^2))(I + tX + O(t^2)) \\ &= I + t(X^\dagger + X) + O(t^2). \end{aligned}$$

Thus $X^\dagger = -X$. In addition,

$$1 = \det A(t) = \det(I + tX + O(t^2)) = 1 + t \operatorname{tr} X + O(t^2),$$

so $\operatorname{tr} X = 0$. Therefore

$$\begin{aligned} \mathfrak{u}(n) &= \{X \in M_n(\mathbb{C}) : X^\dagger = -X\}, \\ \mathfrak{su}(n) &= \{X \in M_n(\mathbb{C}) : X^\dagger = -X, \operatorname{tr} X = 0\}. \end{aligned} \quad (14.11)$$

These are *real* Lie algebras. If $X^\dagger = -X$ and $z \in \mathbb{C}$, then

$$(zX)^\dagger = z^* X^\dagger = -z^* X.$$

This equals $-zX$ for every X only when $z = z^*$, that is, when $z \in \mathbb{R}$. Closure under real scalars, not arbitrary complex scalars, is the reason $\mathfrak{u}(n)$ and $\mathfrak{su}(n)$ are regarded here as real Lie algebras.

14.8.1 A basis and dimension for $\mathfrak{su}(2)$

The three matrices

$$T_1 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \quad T_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad T_3 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$$

are the matrices $i\sigma_1, i\sigma_2, i\sigma_3$, where the σ_i are the Pauli matrices. Each is anti-Hermitian and traceless. Every $X \in \mathfrak{su}(2)$ has the form

$$X = \begin{pmatrix} ia & b \\ -b^* & -ia \end{pmatrix}, \quad a \in \mathbb{R}, \quad b \in \mathbb{C}.$$

Thus $a, \operatorname{Re} b, \operatorname{Im} b$ are three real parameters, and

$$\dim_{\mathbb{R}} \mathfrak{su}(2) = 3.$$

14.8.2 Dimension of $\mathfrak{su}(n)$

An anti-Hermitian matrix is determined by:

- one arbitrary complex entry above the diagonal for each pair $i < j$, contributing $2 \cdot n(n-1)/2 = n(n-1)$ real parameters;
- n purely imaginary diagonal entries.

The trace-zero condition removes one real diagonal parameter. Therefore

$$\dim_{\mathbb{R}} \mathfrak{su}(n) = n(n-1) + (n-1) = n^2 - 1. \quad (14.12)$$

14.9 The special linear algebra

$$\mathfrak{sl}(n, \mathbb{C}) = \{X \in \mathfrak{gl}(n, \mathbb{C}) : \operatorname{tr} X = 0\}$$

is a complex Lie algebra. If X_i are traceless and $\lambda_i \in \mathbb{C}$, then $\sum_i \lambda_i X_i$ is traceless, and

$$\operatorname{tr}[X, Y] = \operatorname{tr}(XY) - \operatorname{tr}(YX) = 0.$$

A convenient complex basis consists of

$$E_{ij} \quad (i \neq j), \quad H_r = E_{rr} - E_{r+1, r+1} \quad (r = 1, \dots, n-1).$$

There are $n(n-1) + (n-1) = n^2 - 1$ basis elements, so

$$\dim_{\mathbb{C}} \mathfrak{sl}(n, \mathbb{C}) = n^2 - 1.$$

Viewed as a real Lie algebra, a real basis is obtained by adjoining iE_{ij} and iH_r , and

$$\dim_{\mathbb{R}} \mathfrak{sl}(n, \mathbb{C}) = 2n^2 - 2.$$

Similarly,

$$\mathfrak{sl}(n, \mathbb{R}) = \{X \in M_n(\mathbb{R}) : \operatorname{tr} X = 0\}, \quad \dim_{\mathbb{R}} \mathfrak{sl}(n, \mathbb{R}) = n^2 - 1.$$

14.10 Important groups and their Lie algebras

Group	Lie algebra	$\dim_{\mathbb{R}}$	$\dim_{\mathbb{C}}$	Matrix condition
$GL(n, \mathbb{R})$	$\mathfrak{gl}(n, \mathbb{R})$	n^2	–	$X_{ij} \in \mathbb{R}$
$GL(n, \mathbb{C})$	$\mathfrak{gl}(n, \mathbb{C})$	$2n^2$	n^2	$X_{ij} \in \mathbb{C}$
$SL(n, \mathbb{R})$	$\mathfrak{sl}(n, \mathbb{R})$	$n^2 - 1$	–	$X \in M_n(\mathbb{R}), \operatorname{tr} X = 0$
$SL(n, \mathbb{C})$	$\mathfrak{sl}(n, \mathbb{C})$	$2n^2 - 2$	$n^2 - 1$	$X \in M_n(\mathbb{C}), \operatorname{tr} X = 0$
$O(n)$	$\mathfrak{o}(n) = \mathfrak{so}(n)$	$\frac{1}{2}n(n-1)$	–	$X \in M_n(\mathbb{R}), X^T = -X$
$SO(n)$	$\mathfrak{so}(n)$	$\frac{1}{2}n(n-1)$	–	same tangent condition as $O(n)$
$U(n)$	$\mathfrak{u}(n)$	n^2	–	$X \in M_n(\mathbb{C}), X^\dagger = -X$
$SU(n)$	$\mathfrak{su}(n)$	$n^2 - 1$	–	$X \in \mathfrak{u}(n), \operatorname{tr} X = 0$
$Sp(2n, \mathbb{R})$	$\mathfrak{sp}(2n, \mathbb{R})$	$n(2n+1)$	–	$X \in M_{2n}(\mathbb{R}), X^T J = -JX$

表 14.1: The important matrix Lie groups and Lie algebras collected in the handwritten lecture. A dash means that the algebra is being treated as real, not as a complex vector space.

14.11 Brackets of subalgebras and the derived algebra

Definition 14.5: Bracket of two subalgebras

If $\mathfrak{h}_1, \mathfrak{h}_2 \leq \mathfrak{g}$, define

$$[\mathfrak{h}_1, \mathfrak{h}_2] = \operatorname{span}_{\mathbb{K}}\{[X_1, X_2] : X_1 \in \mathfrak{h}_1, X_2 \in \mathfrak{h}_2\}.$$

Definition 14.6: Derived Lie algebra

The *derived algebra*, or commutator algebra, is

$$\mathfrak{g}' = [\mathfrak{g}, \mathfrak{g}].$$

Proposition 14.1

The derived algebra $\mathfrak{g}'$ is a Lie subalgebra of $\mathfrak{g}$. In fact, it is an ideal.

Proof. It is a vector subspace by its definition as a span. If $X, Y \in \mathfrak{g}'$, then $X, Y \in \mathfrak{g}$, and the single commutator $[X, Y]$ belongs to the set whose span defines $\mathfrak{g}'$. Hence $[X, Y] \in \mathfrak{g}'$.

For the stronger ideal statement, take $Z \in \mathfrak{g}$ and a generator $[X, Y]$ of $\mathfrak{g}'$. Jacobi gives

$$[Z, [X, Y]] = [[Z, X], Y] + [X, [Z, Y]],$$

which is a sum of commutators and therefore lies in $\mathfrak{g}'$. Linearity extends this to every element of $\mathfrak{g}'$. ■

The notes also express an arbitrary element of the derived algebra in a basis:

$$X = \sum_{i,j} x_{ij} [X_i, X_j], \quad x_{ij} = -x_{ji}.$$

The antisymmetric coefficients suffice because

$$\sum_{i,j} x_{ij} [X_i, X_j] = \sum_{i,j} \frac{x_{ij} - x_{ji}}{2} [X_i, X_j];$$

the symmetric part makes no contribution.

Proposition 14.2: Characterization of Abelian algebras

$$\mathfrak{g} \text{ is Abelian} \iff \mathfrak{g}' = \{0\}.$$

Proof. If $\mathfrak{g}$ is Abelian, every commutator is zero, so its span is zero. Conversely, if $\mathfrak{g}' = \{0\}$, then every $[X, Y] \in \mathfrak{g}'$ is zero. ■

14.12 Structure constants, isomorphisms, and basis changes

14.12.1 Comparing two algebras in chosen bases

Let $\mathfrak{g}, \mathfrak{g}'$ have the same dimension, with bases $\{T_1, \dots, T_d\}$ and $\{T'_1, \dots, T'_d\}$, and structure constants

$$[T_i, T_j] = \sum_k f^k_{ij} T_k, \quad [T'_i, T'_j] = \sum_k f'^k_{ij} T'_k.$$

Define the linear bijection $\psi(T_i) = T'_i$. For

$$X = \sum_i x_i T_i, \quad Y = \sum_j y_j T_j,$$

we find

$$\begin{aligned} \psi([X, Y]) &= \sum_{i,j,k} x_i y_j f^k_{ij} T'_k, \\ [\psi(X), \psi(Y)] &= \sum_{i,j,k} x_i y_j f'^k_{ij} T'_k. \end{aligned}$$

These are equal for all coefficients x_i, y_j if and only if

$$f^k_{ij} = f'^k_{ij} \quad \text{for every } i, j, k.$$

Thus equality of structure constants in matched bases proves that the basis-matching map is a Lie-algebra isomorphism.

14.12.2 How the constants transform under a basis change

Now keep one Lie algebra $\mathfrak{g}$ but change the basis. Let

$$T'_i = \sum_k S_{ki} T_k, \quad S \in GL(d, \mathbb{K}).$$

The inverse relation is

$$T_m = \sum_n (S^{-1})_{nm} T'_n.$$

Then

$$\begin{aligned} [T'_i, T'_j] &= \left[\sum_k S_{ki} T_k, \sum_\ell S_{\ell j} T_\ell \right] \\ &= \sum_{k, \ell} S_{ki} S_{\ell j} [T_k, T_\ell] \\ &= \sum_{k, \ell, m} S_{ki} S_{\ell j} f^m_{k\ell} T_m \\ &= \sum_{k, \ell, m, n} S_{ki} S_{\ell j} f^m_{k\ell} (S^{-1})_{nm} T'_n. \end{aligned}$$

Therefore

$$\boxed{f^m_{ij} = \sum_{k, \ell, m} S_{ki} S_{\ell j} f^m_{k\ell} (S^{-1})_{nm}.} \quad (14.13)$$

There are two factors of S , one for each lower input index, and one factor of S^{-1} for the upper output index. In this precise sense the structure constants transform as the components of a tensor of type $(1, 2)$, or equivalently $(2, 1)$ if one lists the two input slots first. They are basis dependent, but the Lie bracket they describe is intrinsic.

What to retain from this chapter

1. A Lie algebra is a vector space with a bilinear, antisymmetric bracket satisfying the Jacobi identity.

2. In a basis,

$$[T_i, T_j] = f^k_{ij} T_k.$$

Antisymmetry and Jacobi become algebraic constraints on the f^k_{ij} , and conversely these constraints define a bracket.

3. The commutator makes $M_n(\mathbb{K})$ and $\text{End}(V)$ into Lie algebras. Ado's theorem says every finite-dimensional Lie algebra has a faithful matrix realization.

4. The cross-product algebra satisfies

$$(\mathbb{R}^3, \times) \cong \mathfrak{so}(3).$$

5. Differentiating the defining matrix equation at the identity gives the associated matrix Lie algebra:

$$\mathfrak{so}(n) = \{X : X^T = -X\},$$

$$\mathfrak{sp}(2n, \mathbb{R}) = \{X : X^T J + JX = 0\},$$

$$\mathfrak{su}(n) = \{X : X^\dagger = -X, \text{tr } X = 0\}.$$

- 6.

$$\dim \mathfrak{so}(n) = \frac{1}{2}n(n-1), \quad \dim \mathfrak{sp}(2n, \mathbb{R}) = n(2n+1), \quad \dim \mathfrak{su}(n) = n^2 - 1.$$

7. Over $\mathbb{R}$, $\mathfrak{sp}(2, \mathbb{R}) \cong \mathfrak{sl}(2, \mathbb{R})$, not $\mathfrak{so}(3)$; their complexifications are isomorphic.
8. The derived algebra $\mathfrak{g}' = [\mathfrak{g}, \mathfrak{g}]$ is an ideal, and $\mathfrak{g}$ is Abelian exactly when $\mathfrak{g}' = \{0\}$.
9. Structure constants change according to (14.13); the bracket itself does not depend on a basis.